

Note : 1) All questions are compulsory.

2) Figures to the right indicate full marks.

Q.1 Attempt All (Each of 5 marks)

[15M]

a) Multiple Choice Questions

1. _____ command is used to check current working directory.
a) pwd
b) pdw
c) dir
d) file
2. The default symbol for bash shell is _____ sign.
a) #
b)\$
c)%
d) @
3. Vi editor uses _____ command to move the cursor to the left.
a) k
b) j
c) h
d) l
4. _____ OS is not based on Linux.
a) Ubuntu
b) Redhat
c) BCD
d) all of the above
5. _____ is a text editor.
a) GNOME
b) LILO
c) emacs
d) none of the above

b) Fill in the blanks.

(cal, rm, -i, -o, editor, tail)

1. _____ command use to remove files.
2. The vi is nothing but an _____.
3. _____ command is used to print last N number of data of given input.
4. _____ command is use to display calendar.
5. _____ option of ls command is used to view the file with inode number.

c) Short Answers

1. What is half duplex mode in networking?
2. What is the syntax of tar command?
3. What do you mean by shell?
4. How the home directory is represented in Linux?
5. Which command is used to get user information?

Q.2 Attempt the following (Any Three)

[15M]

- a) Explain any five features of Linux operating system.
- b) Explain Linux architecture with proper diagram.
- c) Explain the following-
 - i) Init Scripts of Linux
 - ii) Shutdown Process in Linux
- d) What is importance of Linux in software ecosystem?
- e) Explain Filesystem of Linux.
- f) What is a process? Explain ps and tty command in Linux.

Q.3 Attempt the following (Any Three)

[15M]

- a) Explain following command with suitable example: cp, who, cat, ls, od
- b) What do you mean by Absolute Pathname? Illustrate with example.
- c) Explain following command used for compression of files : bzip2, bzip, gzip, gunzip
- d) What are Command aliases? Why it is use? Explain with example.
- e) Explain use of man and help command in Linux.
- f) Explain three modes of vi editor.

Q.4 Attempt the following (Any Three)

[15M]

- a) Explain the fields of /etc/passwd file.
- b) Write a shell script that will give month number as an argument and will translate this number into month name. (Use Case Statement)
- c) What are Regular Expressions in Linux? Explain any three characters.
- d) Explain awk command with example.
- e) Explain syntax of if...else and if...elif...else command with example.
- f) Explain ssh in detail.

Q.5 Attempt the following (Any Three)

[15M]

- a) What is Linux Kernel? Explain distributions of Linux.
- b) Explain Environment Variables in Linux.
- c) Explain grep command. Give example of it.
- d) Explain Hard Link with example.
- e) Explain all attributes of ls -l command.
